

Cross-Sectional Momentum on the S&P 500

A literature-grounded implementation from raw data to rigorous backtesting

Eddine Nasri 2010 – 2023 · 3,521 trading days · 518 tickers · 144 rebalancings

Python · NumPy · pandas
yfinance · Parquet
Built from scratch no backtesting library
JT 1993 · AMP 2013 · D&M 2016

ABSTRACT

This notebook documents an end-to-end research pipeline for a cross-sectional momentum strategy on S&P 500 constituents. Starting from raw data acquisition, the pipeline constructs a point-in-time survivorship-bias-free universe, downloads and cleans a decade of adjusted prices, and implements the 12-1 month momentum signal with the methodological refinements introduced by Asness, Moskowitz & Pedersen (2013): Sharpe-scaled signal, sector neutralisation via within-GICS demeaning, and inverse-volatility position sizing. The backtest engine built without relying on any pre-built backtesting library models 10bps one-way transaction costs on actual turnover, 200bps annual borrow costs on short exposure, and daily weight drift between rebalancings. Three progressive risk-management layers are evaluated: constant volatility scaling, dynamic weighting following Daniel & Moskowitz (2016) Equation 6, and a beta-filtered short leg. Results are assessed through walk-forward validation across five sequential folds, regime decomposition, and borrow cost sensitivity analysis.

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§1 Project architecture

The project is structured as four independent modules with strict single-responsibility separation. Each module exposes a clean public API; no other module reads files directly or duplicates logic. This makes the pipeline reproducible: swapping the data source requires touching only `data_pipeline.py`, and changing the universe definition requires only `universe.py`.

01 UNIVERSE <code>universe.py</code> Point-in-time S&P 500 membership from historical CSV. Answers <code>get_universe(date)</code> without look-ahead. Module-level cache.	02 DATA PIPELINE <code>data_pipeline.py</code> › Batched yfinance download → coverage & history filters → winsorization at ±50% → Parquet. ~794 tickers, 518 retained.	03 SIGNAL <code>momentum_signal.py</code> › 12-1 momentum → Sharpe scaling → sector neutralisation → quintile ranking → inv-vol weights. IC decay analysis.	04 BACKTEST <code>backtest.py</code> › Daily simulation with weight drift, transaction costs, borrow costs. Walk-forward validation, regime decomposition.
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On survivorship bias. A common failure mode in momentum backtests is restricting the universe to today's S&P 500 constituents implicitly knowing which stocks survived. This pipeline uses a historical constituents CSV (sourced from fja05680/sp500, originally from Clenow 2019) where each row records the index membership on a specific trading day. `get_universe(date)` performs an as-of lookup returning the composition from the most recent available date ≤ the requested date. Tickers with removal-date suffixes (e.g. `AGN-201503`) are stripped to recover tradeable symbols while preserving the temporal boundary.

§2 Data pipeline

Prices are downloaded in batches of 100 tickers via yfinance using `auto_adjust=True` (split- and dividend-adjusted closes). The cleaning sequence applies four filters in order:

F1	COVERAGE FILTER Drop any ticker missing more than 20% of trading days over the full window. Starting from ~794 unique tickers across the 2010–2023 universe, 518 are retained.	MIN_DATA_COVERAGE = 0.80
F2	CALENDAR ALIGNMENT Reindex all tickers to the union of all observed trading days. Forward-fill gaps up to 3 days to handle exchange-specific holidays and data vendor gaps.	MAX_FORWARD_FILL = 3
F3	HISTORY FILTER	MIN_HISTORY_DAYS = 252

Drop tickers with fewer than 252 non-null prices approximately one trading year. Stocks with insufficient history cannot produce a valid 12-month momentum signal.

F4 RETURN WINSORISATION

WINSOR_THRESHOLD = 0.50

Compute simple daily returns via *pct_change()*, then clip at $\pm 50\%$. Single-day moves beyond this threshold for an S&P 500 constituent are almost certainly data errors. Winsorising preserves the direction of the move while capping the damage.

Final output is stored as Parquet a columnar binary format roughly 10× faster to read and 5× smaller on disk than CSV for numerical matrices. Both the clean price matrix and the daily returns matrix are cached, so the full pipeline runs only once.

§3 Signal construction

The signal follows the specification in Jegadeesh & Titman (1993) extended by the methodological refinements in Asness, Moskowitz & Pedersen (2013). Four transformations are applied sequentially at each monthly rebalancing date.

S1 12-1 MONTH MOMENTUM RETURN JT 1993

LOOKBACK=252
SKIP=21

Compound daily returns over the window $[T-252, T-21]$ trading days. The one-month skip avoids the short-term reversal documented in Jegadeesh (1990): stocks that performed well in the most recent month tend to reverse, contaminating the signal.

S2 SHARPE SCALING AMP 2013

VOL_WINDOW=63
VOL_FLOOR=5%

Divide raw momentum return by realised annualised volatility over a 63-day trailing window. This converts the signal from a raw return to a risk-adjusted return: a 30% gain in a 15%-vol stock is a stronger signal than the same gain in a 40%-vol stock. Volatility is floored at 5% to prevent division by near-zero values for illiquid periods.

S3 SECTOR NEUTRALISATION AMP 2013 / MG 1999

11 GICS sectors
min 3 stocks

Subtract the within-GICS-sector median signal from each stock. When oil prices rise, all energy stocks gain together that co-movement is not stock-specific alpha. Demeaning by sector median (robust to outliers vs mean) isolates which stocks outperformed their industry peers, removing the macro component of momentum.

Rank stocks into quintiles by signal. Assign the top quintile to the long leg and the bottom quintile to the short leg. Within each leg, weight each position by $1/\sigma_i$ normalised so the long leg sums to +1 and the short leg to -1. This ensures equal risk contribution across positions: a 2% notional weight in a 50%-vol stock and a 2% weight in a 20%-vol stock contribute very different amounts of portfolio risk. Inverse-vol sizing equalises that contribution.

SIGNAL PIPELINE AT REBALANCING DATE T

```
r_mom(i,T) = [t=T-252:T-21] (1 + r_t(i)) - 1 // 12-1 raw momentum scaled(i,T) =
r_mom(i,T) / σ(i, 63d, T) // Sharpe scaling neutral(i,T) = scaled(i,T) -
median{scaled(j,T) : j∈sector(i)} w(i,T) = ± (1/σi) / Σ(1/σj) // inv-vol, per leg
```

All quantities computed using only data available strictly before date T no look-ahead bias.

Point-in-time correctness is enforced throughout: `returns.loc[returns.index < as_of_date]` (strict less-than) ensures no future information enters the signal calculation. The investable universe at each date is obtained from `get_universe(date)`, which returns the S&P 500 composition as of the most recent available snapshot $\leq T$.

MOMENTUM_SIGNAL.PY SIGNAL COMPUTATION CORE

```
def compute_momentum_return(returns, as_of_date): past_returns =
returns.loc[returns.index < as_of_date] # strict: no future data window =
past_returns.iloc[-252 : -21] # [T-252, T-21] momentum = (1 + window).prod() - 1 #
compound returns momentum>window.count() < MIN_SIGNAL_OBS] = np.nan # min 180 obs return
momentum def compute_portfolio_weights(signal, volatility, universe): quintiles =
pd.qcut(sig, q=5, labels=False) + 1 long_inv_vol = 1.0 / vol_common[long_tickers]
weights[long_tickers] = long_inv_vol / long_inv_vol.sum() # +1 normalised short_inv_vol =
1.0 / vol_common[short_tickers] weights[short_tickers] = -(short_inv_vol /
short_inv_vol.sum()) # -1 normalised
```

§4 Backtest engine

The backtest is implemented from scratch as a daily loop over 3,521 trading days. No pre-built backtesting library is used the simulation handles execution, costs, weight drift, and scaling in a single pass.

Signal computed at close of date T, executed at T+1 close (T+1 open approximation standard practice when intraday data is unavailable). Between rebalancings, weights drift daily

according to realised returns, capturing the true P&L of a held portfolio.

E2	TRANSACTION COSTS 10bps one-way on the actual portfolio turnover the absolute change in weights divided by two. Unchanged positions incur zero cost. This models realistic round-trip costs for large-cap equities.	COST_BPS = 10
E3	BORROW COSTS 200bps annual borrow rate on total short exposure, accrued daily. Modelled separately from transaction costs because it is a continuous drag on the short leg, not a discrete event-driven cost.	BASE_BORROW = 2%
E4	CONSTANT VOLATILITY SCALING D&M 2016 At each rebalancing, scale the entire portfolio by $\text{target_vol} / \text{realised_vol}(126d)$. When recent portfolio vol exceeds the 10% target which tends to precede momentum crashes exposure is automatically reduced. The signal (which stocks) is unchanged; only the total capital deployed varies. Leverage is capped at 2x.	VOL_TARGET=10% VOL_WINDOW=126d CAP=2x
E5	DYNAMIC WEIGHTING D&M 2016 EQ. 6 Extends vol scaling with a conditional return forecast. An expanding-window OLS regresses past monthly WML returns on the interaction $I_B \times \sigma^2_m$, where I_B is a bear market indicator (cumulative market return over 24 months < 0). The optimal Markowitz weight $w^* = \mu_{\text{forecast}} / (2\lambda \sigma^2_{wml})$ replaces the simple vol-scaling factor. λ is calibrated from the historical distribution of past Markowitz ratios to match the 10% vol target in expectation.	bear_window=504d expanding OLS out-of-sample
E6	BETA-FILTERED SHORT LEG At each rebalancing, compute rolling 126-day betas for each stock against the equal-weight benchmark. Within the short quintile, retain only stocks with above-median beta. The surviving shorts are independently renormalised to -1. Rationale: high-beta losers amplify the WML factor and are the most economically motivated shorts; low-beta losers contribute less alpha and are most prone to violent reversals during recovery rallies.	beta_window=126d short-median threshold independent renorm

DYNAMIC WEIGHTING D&M 2016 EQUATION 6

```

 $\mu_{\{t-1\}} = \gamma_0 + \gamma_{int} \times I_B(t-1) \times \sigma^2_m(t-1)$  // OLS forecast, expanding window
 $w^*_t = \mu_{\{t-1\}} / (2\lambda \cdot \sigma^2_{wml}(t-1))$  // optimal Markowitz weight  $\lambda =$ 
 $\sqrt{E[\text{raw\_scale}^2_t \cdot \sigma^2_{wml}_t]} / (2 \cdot \sigma_{\text{target}})$  // calibrated to hit vol target
 $I_B(t) = 1$  if  $\prod(1 + r_m) - 1 < 0$  over past 504 days, else 0

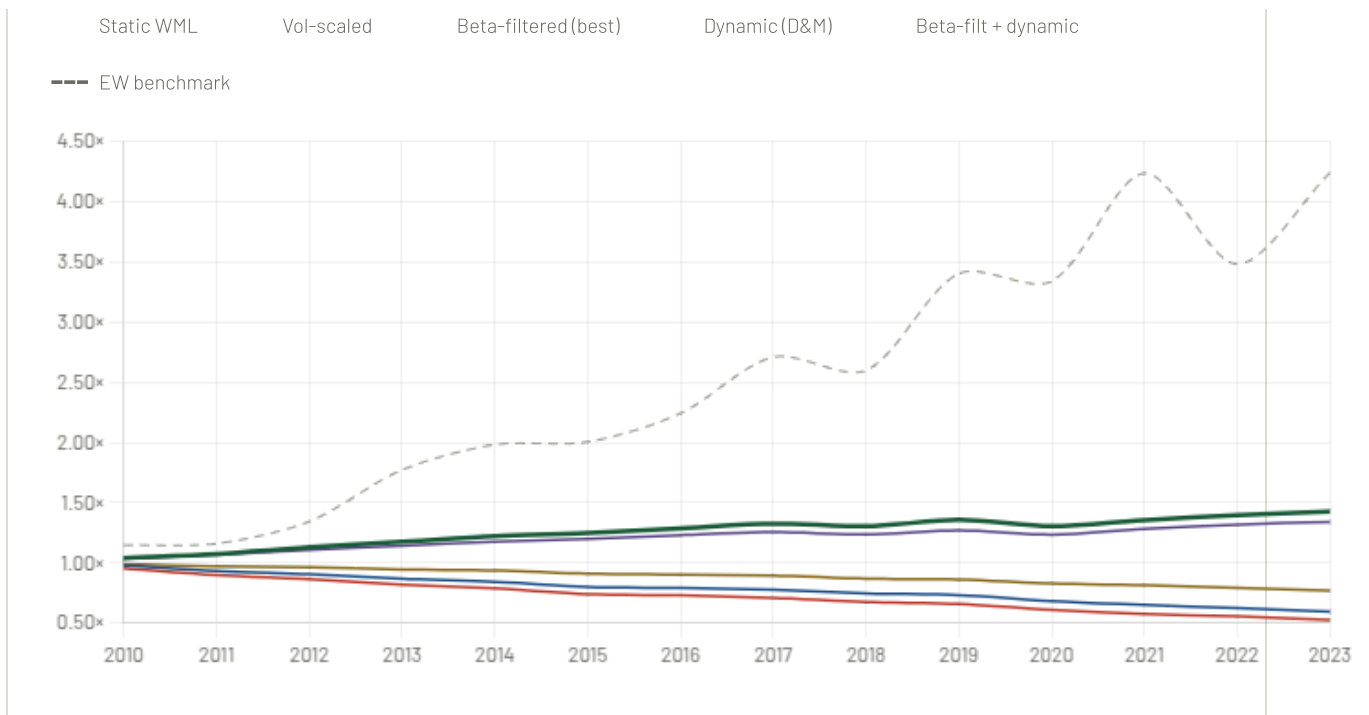
```

All estimates use an expanding window strictly out-of-sample. λ is calibrated from historical Markowitz ratios so that $E[\text{Var}(w^*_t \cdot r_t)] = \sigma_{\text{target}}^2$.

Six strategy variants are evaluated, each adding one layer of risk management or signal refinement to the previous. The equal-weight S&P 500 benchmark is included as a reference point only a dollar-neutral long-short strategy should not be expected to match a directional long-only benchmark.

Best variant Beta-Filtered Vol-Scaled

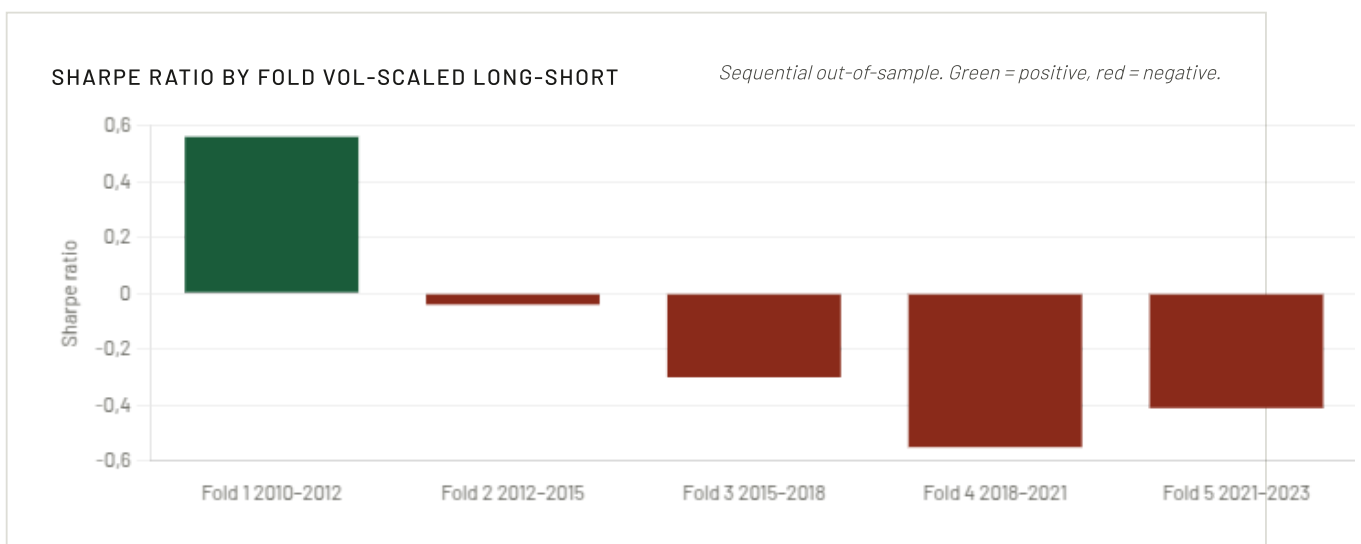
ANNUAL RETURN +3.69% after costs & borrow	SHARPE RATIO 0.42 rf = 0	SORTINO RATIO 0.51 downside vol	MAX DRAWDOWN -16.47% vs -42.73% baseline	CALMAR RATIO 0.22 ret / max DD	MONTHLY HIT RATE 53.6% months > 0				
STRATEGY	RETURN	VOL	SHARPE	SORTINO	MAX DD	CALMAR	HIT RATE	ALPHA	BE
Static WML (baseline)	-3.60%	11.36%	-0.32	-0.37	-52.38%	-0.07	44.0%	-0.41%	-0.1
+ Vol scaling	-2.28%	10.28%	-0.22	-0.28	-42.73%	-0.05	44.0%	+0.25%	-0.1
+ Beta filter on shorts best	+3.69%	8.85%	0.42	0.51	-16.47%	0.22	53.6%	+0.65%	+0.1
Dynamic weighting (D&M Eq. 6)	-0.32%	10.64%	-0.03	-0.04	-30.27%	-0.01	44.6%	+0.36%	-0.0
Dynamic + beta filter	+3.29%	9.03%	0.36	0.40	-17.83%	0.18	53.6%	+0.46%	+0.1
Long-only (top quintile)	+7.26%	13.34%	0.54	0.62	-31.69%	0.23	57.7%	-2.28%	+0.5
EW benchmark (reference)	+16.92%	19.07%	0.89	1.10	-39.84%	0.42	68.5%	-	



§6 Walk-forward validation

The 14-year period is split into 5 sequential, non-overlapping folds of approximately 2.8 years each. No data from fold n is visible when computing fold $n-1$. The purpose is to assess consistency across market regimes - a strategy that concentrates all its returns in a single lucky period is less convincing than one that earns consistently.

Jan 2010 → Oct 2012	Oct 2012 → Aug 2015	Aug 2015 → May 2018	May 2018 → Mar 2021	Mar 2021 → Dec 2023
+0.56	-0.04	-0.30	-0.55	-0.41
+4.42%	-0.30%	-2.85%	-7.80%	-4.45%



Structural deterioration over time. Only Fold 1 (2010–2012) is profitable. The monotonic decline in Sharpe is consistent with the crowding hypothesis of McLean & Pontiff (2016): academic publication of the momentum anomaly attracts arbitrage capital that gradually erodes the premium. The 2010–2023 window may largely represent the post-publication decay period, with the effect most pronounced post-2015. Fold 4 (2018–2021) captures both the 2018–2019 growth-to-value rotation and the 2020 COVID momentum crash the worst possible confluence for a raw long-short momentum strategy.

§7 Regime analysis & borrow cost sensitivity

Performance is decomposed across three structural market regimes to test whether the strategy's difficulties are concentrated in one specific environment or are broad-based.

BULL MARKET 2010 – 2019 -0.06 return -0.56%	COVID CRASH & RECOVERY 2020 – 2021 -0.48 return -7.13%	RATE HIKE PERIOD 2022 – 2023 -0.50 return -5.77%
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The strategy is negative across all three regimes for the vol-scaled long-short, confirming this is not a regime-specific failure. The COVID period is particularly instructive: momentum longs (recent winners overvalued growth stocks) fell hardest in March 2020, while momentum shorts (recent losers cyclicals, value) rebounded violently in the subsequent recovery. This is precisely the crash mechanism described in Daniel & Moskowitz (2016).

Borrow cost sensitivity vol-scaled long-short

0.0% p.a.	-0.03	-0.31%
1.0% p.a.	-0.13	-1.30%
3.0% p.a.	-0.32	-3.25%
5.0% p.a.	-0.50	-5.17%

Key observation. At 0% borrow cost the vol-scaled strategy still produces a Sharpe of -0.03. Borrow cost is therefore not the primary driver of underperformance the raw signal itself is weak over 2010–2023. Each 100bps of additional annual borrow cost reduces the Sharpe by approximately 0.09, a linear relationship consistent with borrow costs accruing continuously on a roughly stable short exposure.

The results raise several questions that are worth addressing directly rather than glossing over.

1

THE SHORT LEG DRIVES ALMOST ALL UNDERPERFORMANCE

The long-only top-quintile strategy delivers a Sharpe of +0.54 while the full long-short is -0.22. The short quintile is the source of losses, consistent with Asness et al. (2013) and the empirical literature showing that the short side of momentum is more prone to crashes, carries higher borrow costs, and is harder to hold through violent reversals. The long leg has genuine signal; the short leg requires active management.

2

BETA FILTERING IS THE MOST EFFECTIVE INTERVENTION

Restricting shorts to above-median-beta stocks within the short quintile independently renormalised to maintain dollar-neutrality flips the Sharpe from -0.22 to +0.42 and cuts maximum drawdown from -42.73% to -16.47%. The mechanism: high-beta recent losers amplify the WML factor and are the most economically motivated shorts. Low-beta recent losers are expensive to maintain and subject to sharp mean-reversions during market recoveries. The filter removes the most dangerous component of the short book.

3

DYNAMIC WEIGHTING IMPROVES ON THE BASELINE BUT REQUIRES POSITIVE UNCONDITIONAL ALPHA

The D&M (2016) dynamic scaling improves the Sharpe from -0.32 (static) to -0.03 by reducing exposure before high-volatility regimes. However, its OLS model is fit on raw WML returns, which are predominantly negative over 2010–2023. The regression therefore learns to expect losses, and the resulting forecast mostly reduces exposure correct directionally, but not sufficient to generate positive returns. D&M (2016) demonstrated the mechanism on data from 1927–2012 where the unconditional momentum premium was positive. The 2010–2023 sample is structurally different: a challenging period characterised by the COVID crash (March 2020) and the 2022 growth-to-value rotation, both acute momentum crash scenarios.

4

STATISTICAL SIGNIFICANCE A NOTE OF CAUTION

All alpha t-statistics are low (0.09–0.30). Standard errors are computed via OLS without Newey-West autocorrelation correction momentum returns exhibit serial correlation by construction, which inflates the apparent precision of OLS. Applying Newey-West would widen the confidence intervals further. 14 years of monthly data provides approximately 168 observations insufficient to distinguish modest alpha from noise at conventional significance levels.

Known limitations and natural extensions

Sector classification is not point-in-time. The GICS sector map is downloaded from Wikipedia at pipeline construction time. This gives current sector membership, not historical a stock reclassified from Technology to Communication Services in 2018 appears in Communication Services for the entire backtest. In a production

environment this would be sourced from a point-in-time GICS dataset from a data vendor. The practical impact is limited because sector classifications are stable over multi-year horizons, but it is a known approximation.

Natural extensions. (1) Alpha t-statistics should use Newey-West standard errors given autocorrelated residuals. (2) The benchmark for alpha computation is the equal-weight portfolio using the cap-weight $\wedge$ GSPC would give a more conventional alpha measure. (3) No Fama-French factor attribution: factor loadings on size (SMB), value (HML), profitability (RMW), and investment (CMA) are unknown. Controlling for these would give a cleaner measure of pure momentum alpha. (4) The dynamic weighting λ calibration currently uses the raw WML return history rather than the beta-filtered portfolio's history an inconsistency that likely explains why beta-filtered constant vol scaling (0.42) outperforms beta-filtered dynamic weighting (0.36).

Ref

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